A* Examination Guide

Built from 16 mark schemes and 14 question papers, January 2019 to June 2026

This guide is built from the Paper 2 question papers and mark schemes. Model answers follow the marking points examiners list, and every warning comes from a “reject”, “ignore”, “allow” or “max” note. Paper 2 covers the whole specification, but leans towards applied contexts: teeth, skin, drugs and medicines, disease control, genetic conditions, hormones in reproduction and data-based investigations.

1 How Paper 2 is built
2 Twenty mark-scheme rules
3 Cells, tissues and transport
4 Molecules, diet and food tests
5 Digestion and teeth
6 Blood, heart and vessels
7 Breathing and respiration
8 Excretion and skin
9 Coordination: nerves, brain, eye, ear
10 Hormones, homeostasis and reproduction
11 Genetics and genetic conditions
12 Disease and immunity
13 Drugs, medicines and lifestyle
14 Practical skills and calculations
15 Final A* checklist

Papers analysed: Jan 2019 (legacy 4HB0), Jun 2019, Jan 2020, Nov 2020, Jun 2021, Nov 2021, Jan 2022, Jun 2022, Jan 2023, Jun 2023, Nov 2023, Jun 2024, Nov 2024, Jun 2025, Nov 2025, Jun 2026. Papers are cited by series, e.g. (Jun 25).

1 HOW PAPER 2 IS BUILT

Format: 1 hour 45 minutes, 90 marks, 7–9 structured questions: about 1.2 minutes per mark. The front page says “Show all the steps in any calculations and state the units”. Expect a short labelling or matching question first, then data analysis, a graph, a calculation and one or two 5–6 mark explanations or methods. Some recent papers include a short passage to read and answer from (Nov 25).

Topic	4HB1 papers (of 15)	What is asked most
Calculations and data	15	magnification, percentage change, ratio, rate, scaling to a population, units
Genetics and mutations	13	mutation → protein, sex-linked and dominant conditions, probability with sex
Disease, immunity, antibiotics	12	pathogen table, active/passive immunity, resistance, aseptic technique
Nerves, brain, senses	12	synapse, brain-part functions, pupil, accommodation, 3D vision, ear
Blood, heart and vessels	11	vessel structure, red cells, clotting, tissue fluid, heart defects
Hormones and reproduction	9	FSH, LH, oxytocin, IVF, contraception, glucose control
Breathing and spirometry	8	trace reading (rate, tidal volume), gas exchange, smoking
Drugs and lifestyle	8	alcohol units, addiction, statins, codeine, ACE inhibitors
Diet and food tests	8	Benedict’s, biuret, emulsion, DCPIP for vitamin C, nutrient roles
Kidney, urea, ADH	8	deamination, ultrafiltration, concentrated urine, transplants
Teeth	4	types, decay mechanism, molars and crevices, prevention

Topic	4HB1 papers (of 15)	What is asked most
Skin	4	layers, sweating, vasodilation, functions
Osmosis and diffusion practicals	5	potato cylinders, agar cubes, SA:V

2 TWENTY MARK-SCHEME RULES

1. **"Ignore amount":** name the variable precisely – volume, mass, concentration, type, temperature, time.
2. **Enzymes are denatured;** "reject dies". Explain through the active site and enzyme–substrate complexes.
3. **Osmosis:** water moves from high to low water potential through a partially permeable membrane; "ignore references to concentrations of water".
4. **Immunity wording:** "do not allow destroy/kill/attack disease" (Jan 22) and "ignore fight disease" (Nov 20). Say antibodies bind to antigens on the pathogen; phagocytes engulf it.
5. **Placenta, not umbilical cord,** for antibodies or drugs reaching the fetus ("ignore umbilical cord", Nov 23).
6. **Teeth:** molars have ridges/crevices/grooves; "ignore cusps" (Jun 19, Jan 20).
7. **Reflex actions** are rapid, automatic and protective; "ignore sub-conscious/unconscious" (Jun 25).
8. **Respiration in words when asked:** glucose + oxygen → carbon dioxide + water; "reject formulae" there (Jun 23).
9. **Spell organ names correctly:** "spelling must be correct" for labels such as oviduct and prostate (Jun 23).
10. **Tissue examples are types,** not single items: "reject 'a bone', 'a muscle'" (Nov 23).
11. **Tables and ticks:** one tick per row or column unless told otherwise; extra ticks lose marks.
12. **Thymine, not thiamine** (Jun 24); uracil in RNA.
13. **Genetic diagrams:** genotypes, gametes, offspring, phenotypes; a Punnett square without labels scores at most 2.
14. **Probabilities with sex:** multiply, e.g. boy (0.5) × sickle cell (0.25) = 0.125 (Nov 23).
15. **Calculations:** show working (error carried forward); give the s.f. or form asked (22 250 → 22 300 to 3 s.f., Nov 24; standard form 3.1×10^3, Nov 25).
16. **Graphs:** linear scale using at least half the grid, labelled axes with units, accurate plots (±½ square), lines as instructed; a key for two lines.
17. **Describe data:** trend + figures with units + exceptions; compare groups with comparative words.
18. **Reliability:** repeat and calculate a mean; identify anomalies; use more people/samples (Nov 25: "ignore accuracy, validity" for the reliability mark).
19. **Safety needs a hazard and a precaution** (broken glass – sweep up; biuret irritant – goggles); "ignore wear gloves" in Jun 19.
20. **Link explanations:** each statement followed by "so" or "because"; one point per mark plus a spare.

3 CELLS, TISSUES AND TRANSPORT

DEFINITIONS

Tissue: a group of similar cells working together to carry out a particular function (e.g. ciliated epithelium, squamous epithelium, muscle, nervous, blood, bone).

Organ: a group of different tissues working together for a particular function.

Electron vs light microscope: greater magnification and resolution, so small organelles (ribosomes, ER, mitochondria) are visible (Jun 19, Jun 26).

OSMOSIS PRACTICAL · JUN 25, JUN 26

Cut equal potato cylinders (same size, same potato), blot and weigh; place in a range of sugar or sucrose concentrations for the same time and temperature; blot dry and reweigh; calculate percentage change in mass.

In dilute solutions water enters by osmosis (high to low water potential), so mass increases; in concentrated solutions it falls. Where the line crosses 0% change the water potentials are equal (about 0.45 au in Jun 25).

Blot because surface liquid adds mass and makes readings inaccurate. 7.0 g × 1.04 = 7.3 g after a 4% gain.

- Diffusion in agar cubes (Jan 23): a 2 mm cube has SA 24 mm^2, volume 8 mm^3, SA:V 3 : 1; smaller cubes have shorter diffusion distances, so they decolourise faster.
- Active transport: against the concentration gradient, using ATP from respiration (Jun 26).
- Mitochondria: aerobic respiration, ATP; ribosomes: protein synthesis.

4 MOLECULES, DIET AND FOOD TESTS

Test	Method	Positive result
reducing sugar	Benedict's; heat in a water bath (≥ 60 °C)	blue → green/yellow/orange/brick red
protein	biuret solution	blue → purple/lilac
starch	iodine solution	orange-brown → blue-black
lipid	chop, add ethanol, shake, add water	milky white emulsion
vitamin C	add juice drop by drop to DCPIP	blue DCPIP decolourises; fewer drops = more vitamin C

- Molecule matching (Nov 25, Jun 26): DNA – in chromosomes; maltose – reducing sugar; lipid – contains glycerol; glycogen – storage carbohydrate in the liver; starch – glucose units; pepsin – enzyme; glucagon – raises blood glucose.
- Balanced diet (Nov 24): all nutrients in the correct proportions. Bones (Jun 23): calcium for compact bone, vitamin D for calcium absorption, protein for growth.
- DCPIP method (Jun 22): same volume of DCPIP each time; record volume of juice needed; repeat with boiled juice and compare (heat destroys vitamin C).
- Serving calculations (Nov 24): 11 800 kJ ÷ 118 kJ per slice ≈ 100 slices; 2.4 g per 100 g × 25 = 60 g.

5 DIGESTION AND TEETH

MODEL ANSWER · TOOTH DECAY (4) · NOV 21, JUN 23

Sugar coats the teeth and forms plaque; bacteria in plaque respire the sugar and produce lactic acid; the acid dissolves the enamel and then the dentine. Molars decay most because their ridges and crevices trap food and are hard to reach with a brush. Prevention: brush with fluoride toothpaste, floss, fewer sugary foods, visit the dentist, fluoridated water.

- Teeth (Jan 20): incisors cut, canines tear, premolars and molars grind. Adults have premolars and wisdom teeth; tooth layers: enamel, dentine, pulp cavity.
- Bile (Jun 25): made in the liver, stored in the gall bladder; emulsifies fat into small droplets, increasing the surface area for lipase; neutralises stomach acid.
- Pancreatic juice (Jun 25): protease → amino acids, amylase → maltose, lipase → fatty acids and glycerol, released into the duodenum.
- No bile (Jun 19, 5 marks): less emulsification, lower surface area, less fat digested; small intestine stays acidic so enzymes work less well; more fat egested.

- Lipase and pH (Nov 23): fatty acids from fat digestion lower the pH; without bile salts there is no emulsification, so digestion is slower.

6 BLOOD, HEART AND VESSELS

Vessel	Structure	Reason
artery	thick muscular and elastic wall, narrow lumen, no valves	withstands and maintains high pressure
vein	thin wall, wide lumen, valves	low pressure; valves prevent backflow
capillary	wall one cell thick, pores, narrow lumen	short diffusion distance; cells pass in single file

MODEL ANSWER · RED BLOOD CELL ADAPTATIONS (4) · NOV 21

Haemoglobin combines with oxygen; no nucleus, so more room for haemoglobin; biconcave shape gives a larger surface area; small and flexible to pass through capillaries; thin membrane shortens the diffusion distance.

- Clotting (Jan 23): fibrinogen is converted to fibrin, which forms the mesh; without it no clot forms.
- Tissue fluid (Jan 23): high pressure at the arterial end forces water and small molecules through pores; proteins and red cells stay in; excess drains into lymph, moved by muscle contraction.
- Antibodies (Jan 23): each has a variable region with a shape complementary to one antigen, so it binds only that.
- Hole between ventricles (Jan 19): oxygenated and deoxygenated blood mix, less oxygen reaches cells, less aerobic respiration.
- Circulation order (Jun 23): vena cava → right atrium → pulmonary artery → lungs → pulmonary vein → left side → aorta → organs (e.g. renal artery).

7 BREATHING AND RESPIRATION

READING A SPIROMETER TRACE · JUN 21, JUN 24, JUN 26

Breathing rate: count peaks in a measured time and scale to one minute (10 breaths in 60 s = 10 per minute).

Tidal volume: height of one normal peak converted with the scale (6 squares ÷ 4 squares per dm^3 = 1.5 dm^3).

After exercise: peaks are higher (greater depth) and closer together (higher rate). Method: nose clip, breathe through the mouthpiece, soda lime absorbs CO_2, pen traces on a rotating drum; sterilise the mouthpiece.

- Gas exchange in alveoli: many alveoli (large surface area), one-cell-thick walls, moist lining, dense capillaries.
- Trachea (Jan 22, Nov 25): ciliated epithelium and goblet cells; mucus traps pathogens and dust, cilia sweep it up; cartilage rings keep the airway open. Smoking paralyses cilia and increases mucus.
- Smoking in pregnancy (4, Jan 22): CO combines with haemoglobin (carboxyhaemoglobin), less oxygen reaches the fetus, less respiration, lower birth mass.
- Altitude training (5, Jan 23): more red blood cells and haemoglobin, more oxygen carried to muscles for aerobic respiration, more energy released.
- Respiration: glucose + oxygen → carbon dioxide + water, in mitochondria; ATP → ADP + Pi releases energy.
- Metabolic rate in cold weather (Jun 23): higher, because more respiration releases heat to maintain body temperature.

8 EXCRETION AND SKIN

UREA AND THE KIDNEY

Urea is made in the liver by deamination of excess amino acids (amine group removed). It is used because it has low toxicity, is very soluble and needs little water to remove (Jan 19).

Ultrafiltration: high pressure in the glomerulus forces small molecules into the Bowman's capsule; proteins and blood cells stay. Renal vein vs artery: less urea, less oxygen and glucose, more CO_2.

Concentrated urine (5, Jan 23): less water in the blood is detected by the hypothalamus; ADH is released; the collecting duct becomes more permeable; more water is reabsorbed; a smaller volume of concentrated urine.

MODEL ANSWER · SKIN IN HOT CONDITIONS (6) · JAN 22, JUN 26

Sweat glands secrete more sweat onto the skin surface; the sweat evaporates, taking heat away; arterioles dilate (vasodilation), bringing more blood near the surface, so the skin looks red; heat is lost by radiation. Hair erector muscles relax so hairs lie flat.

Skin layers: epidermis, dermis, subcutaneous fat. Functions: barrier to pathogens and UV, sensory reception, excretion of sweat. Melanin absorbs UV, lowering the risk of skin cancer (Jun 21).

- Diabetes and urine (Jun 23): glucose in the filtrate lowers its water potential, so less water is reabsorbed and more urine is produced.
- Protein in urine (Jun 25) means the glomerulus or basement membrane is damaged: protein is too large to pass normally.

9 COORDINATION: NERVES, BRAIN, EYE, EAR

Brain part	**Function (Jan 20, Jun 21)**
cerebrum (cerebral hemispheres)	intelligence, thought, language, voluntary movement
cerebellum	muscle tone, posture, balance, coordination
medulla	heart rate and breathing
hypothalamus	body temperature, osmoregulation
pituitary gland	releases ADH, FSH, LH, oxytocin

MODEL ANSWERS · SYNAPSE (6) AND ONE-WAY TRANSMISSION (4) · JAN 22, NOV 24

Vesicles containing neurotransmitter move to and fuse with the pre-synaptic membrane; neurotransmitter is released and diffuses across the synaptic cleft; it binds to receptors on the post-synaptic membrane; a new impulse starts.

One direction only: neurotransmitter is released only from the pre-synaptic side and receptors are only on the post-synaptic membrane.

	Nervous system	**Hormonal system (Jan 22)**
carried by	nerve cells	blood
signal	electrical	chemical
speed	fast	slow

- Pupil in dim light (5, Jun 21): retina detects less light; radial muscles of the iris contract, circular muscles relax; pupil widens; more light enters. In bright light the reflex protects the retina.
- Near object (Jan 22): ciliary muscle contracted, suspensory ligaments loose, lens thick. Distant: relaxed, stretched, thin.

- Stereoscopic vision: two eyes give slightly different views, improving judgement of distance. Method (6): catch a ball thrown a fixed distance 10 times with both eyes and with one eye patched; same ball and lighting; compare.
- Retina and optic nerve (Jun 25): rods and cones convert light into impulses; the optic nerve carries them to the brain. Fovea has most cones.
- Ear (Nov 20, Jun 22): pinna collects sound; ossicles transmit and amplify vibrations from eardrum to oval window; cochlea generates impulses; Eustachian tube equalises pressure on both sides of the eardrum.
- Parkinson's (Jan 20): less dopamine (a neurotransmitter), fewer impulses to muscles, tremor and slow movement.
- Reflex arc (Jun 22, Jun 25): receptor → sensory → relay → motor → effector; synapses ensure one direction.

10 HORMONES, HOMEOSTASIS AND REPRODUCTION

MODEL ANSWER · BLOOD GLUCOSE (5) · NOV 25

Blood glucose rises; this is detected by the pancreas (islets of Langerhans); insulin is secreted into the blood plasma; liver cells convert glucose to glycogen; glucose level falls; negative feedback stops insulin secretion. If glucose falls, glucagon causes glycogen to be converted back to glucose.

Hormone	Effect
FSH	follicle development; stimulates oestrogen
oestrogen	repairs the uterus lining
LH	triggers ovulation; maintains the corpus luteum, which makes progesterone
progesterone	maintains the uterus lining
oxytocin	uterus contractions in labour; milk release; bonding (Jun 25)

- IVF (4, Jun 19): FSH and LH given; eggs removed from the ovary; sperm collected; eggs fertilised outside the body; embryos formed and placed in the uterus.
- Identical twins (Jun 19): one zygote splits and each half develops.
- Contraception (Nov 24): barrier (condom, cap), hormonal (pill), rhythm, sterilisation, IUD.
- Sperm (Nov 23): acrosome releases enzymes to digest the ovum's outer layer so the nucleus can enter; mitochondria supply ATP for swimming.

11 GENETICS AND GENETIC CONDITIONS

MODEL ANSWER · MUTATION (3–4) · EVERY SERIES

A change in the base (nucleotide/codon) sequence of DNA; a different amino acid sequence; a different protein with a different shape (e.g. an enzyme that no longer binds its substrate); a changed phenotype.

Condition	Inheritance	Key point
sickle cell (Jun 23, Nov 23)	recessive, autosomal	distorted red cells, less oxygen carried
Huntington's (Jan 22)	dominant	affects both sexes equally; one allele enough
haemophilia (Nov 20, Jun 24)	recessive, X-linked	males affected (one X); fibrin not formed
Duchenne (Jun 22)	recessive, X-linked	carrier females; male cannot carry it on Y
PKU (Jan 20)	recessive	low-protein diet limits phenylalanine
Down's syndrome (Nov 25)	chromosome mutation	three copies of chromosome 21

- Probabilities: boy with sickle cell from Aa × Aa = 0.5 × 0.25 = 0.125 (Nov 23); two haemophiliac sons in a row = ¼ × ¼ = 1/16 (Nov 20); boy with dominant condition from Hh × hh = 0.5 × 0.5 = 0.25 (Jun 21).
- Blood groups (Jun 25): $I^BI^o \times I^BI^o$ gives group B : O = 3 : 1 (75% group B).
- Patient's own stem cells (Jan 23): no embryos destroyed, same antigens so less rejection; risks: tumours, infection, unlicensed clinics.
- Golden rice (Jan 23): restriction enzymes cut out the beta-carotene gene, insert it into a plasmid, transfer into rice cells.
- Birth defects and maternal age (Nov 25): 15 000 × 22 ÷ 100 000 = 3.3 babies.

12 DISEASE AND IMMUNITY

Disease	Pathogen	Transmission / control
malaria	protozoan	mosquito vector; nets, insecticide, drain water
cholera, typhoid	bacterium	contaminated water/food, houseflies; clean water, sewage treatment
polio	virus	droplets and water; vaccination
AIDS	virus (HIV)	sexual intercourse; barrier methods
athlete's foot, thrush	fungus	contact; antibiotics do not work on fungi
tuberculosis	bacterium	airborne droplets

MODEL ANSWERS · IMMUNITY · JAN 22, NOV 23

Active: the body makes its own antibodies after infection or vaccination (weakened/dead pathogen); memory cells give a faster secondary response. **Passive:** antibodies are given (injection, placenta, breast milk); immediate but short-lived.

Booster (Nov 23): antibody level falls below the protective level over time; a booster raises it faster and higher and keeps it high for longer.

MODEL ANSWERS · MONOCLONAL ANTIBODIES (5) AND VIRUSES (6) · JAN 20, JUN 22

Antigen injected into a mouse; spleen lymphocytes make antibodies; they are fused with myeloma (tumour) cells to form hybridoma cells, which divide and produce monoclonal antibodies.

Virus replication: attaches to a host cell; injects its DNA; the DNA is incorporated and replicated; the cell makes viral proteins; new viruses assemble; the cell bursts (lysis) and releases them.

- Resistance (4, Jun 24): mutation; resistant bacteria survive; they reproduce and pass on the resistance allele; allele frequency rises. MRSA is resistant to many antibiotics (Jan 20).
- Antibiotic test (Jun 24): sterilise equipment, flame the loop, sterile agar, lift the lid minimally; the largest clear zone shows the most effective antibiotic.
- Antibiotics do not work on viruses (Ebola, Nov 21) or fungi. Bactericidal antibiotics kill bacteria; others only stop growth.
- Convalescent plasma (Nov 20): antibodies from a recovered patient bind to the virus antigens; plasma has no red cells, so no agglutination.
- Decomposers (Jun 24): release enzymes that break down dead matter and return nutrients to the soil.

13 DRUGS, MEDICINES AND LIFESTYLE

- Alcohol (Jun 19, Jun 24): a depressant; less neurotransmitter released and bound, so slower impulses and reactions; liver cirrhosis and less bile; reduces ADH, so more dilute urine.
- Units: 250 cm^3 of 14% wine contains 35 cm^3 alcohol = 3.5 units (1 unit = 10 cm^3).
- Addiction and withdrawal (Nov 24, Jun 26): compulsive use despite harm; withdrawal symptoms (sweating, nausea, anxiety) when stopped. Codeine is converted to morphine by an enzyme whose activity varies between people.
- Medicines in pregnancy (Nov 23, Jun 26): drugs cross the placenta into the fetus or pass into breast milk; liver or kidney disease means drugs are not broken down or excreted.
- Statins (Nov 23): symptoms were similar with a placebo, so statins may not be the cause; consider age and lifestyle.
- ACE inhibitors (Jun 25): block the enzyme forming angiotensin II, so arterioles do not constrict and blood pressure falls.
- Non-communicable diseases (Nov 24): risk factors – diet, exercise, smoking, alcohol, stress; more common as people live longer.
- Osteoporosis (Jan 23): lower bone density, brittle bones, fractures, loss of height.

14 PRACTICAL SKILLS AND CALCULATIONS

Calculation	Past-paper example
magnification	60 mm ÷ 0.05 mm = ×1200 (Jun 19); 122 000 µm ÷ 50 µm = ×2440 (Nov 23)
standard form	31 000 µm ÷ 10 µm = 3.1×10^3 (Nov 25)
actual size	44 mm ÷ 650 = 0.068 mm (Jan 22)
percentage change	(14 − 20) ÷ 20 × 100 = −30% (Jun 22); (715 − 75) ÷ 75 × 100 = 853% (Nov 20)
percentage of a total	13 ÷ 30 × 100 = 43.3% (Nov 21)
rate	710 cm^3 ÷ 4 h = 178 cm^3/h (Jan 22); 520 ÷ 5 = 104, given as 100 AU/h (Nov 21)
ratio	445 : 185 = 2.41 : 1 (Nov 24); 2.75 : 2.6 = 1.06 : 1 (Jun 22)
scaling to a population	5 000 000 ÷ 100 000 × 445 = 22 250 → 22 300 (Nov 24)
percentage of a volume	4600 × 0.7 = 3220 cm^3 (Jun 24); 4% of 8.5 dm^3 = 0.34 dm^3 (Jan 22)
SA:V	2 mm cube: 24 : 8 = 3 : 1 (Jan 23)

WRITING A METHOD (4–7 MARKS)

State what you change (with a range), what you measure and how (instrument, units), at least two named control variables, repeats and a mean, and a safety point with its reason. Results tables: two columns, headings with units, independent variable first.

Pulse and exercise (Jun 23): resting pulse at wrist or neck for 60 s; exercise for a set time and intensity; measure immediately; rest until the pulse returns to normal; repeat; same person and conditions.

15 FINAL A* CHECKLIST

1. Count the marks and give one extra creditworthy point.
2. Use the exact terms: denatured, active site, water potential, antigen/antibody, placenta, ridges/crevices.
3. Name variables precisely; never “amount”.
4. Data: describe the trend, quote figures with units, then explain with biology.
5. Calculations: show steps, convert units, give the s.f., form and units asked.

6. Genetics: complete diagrams; multiply probabilities when sex is involved.
7. Methods: change, measure, control, repeat, mean, safety with reason.
8. Explain each point with “so” or “because”.
9. Check spelling of organ and structure names.
10. Leave 10 minutes to recheck calculations and long answers.